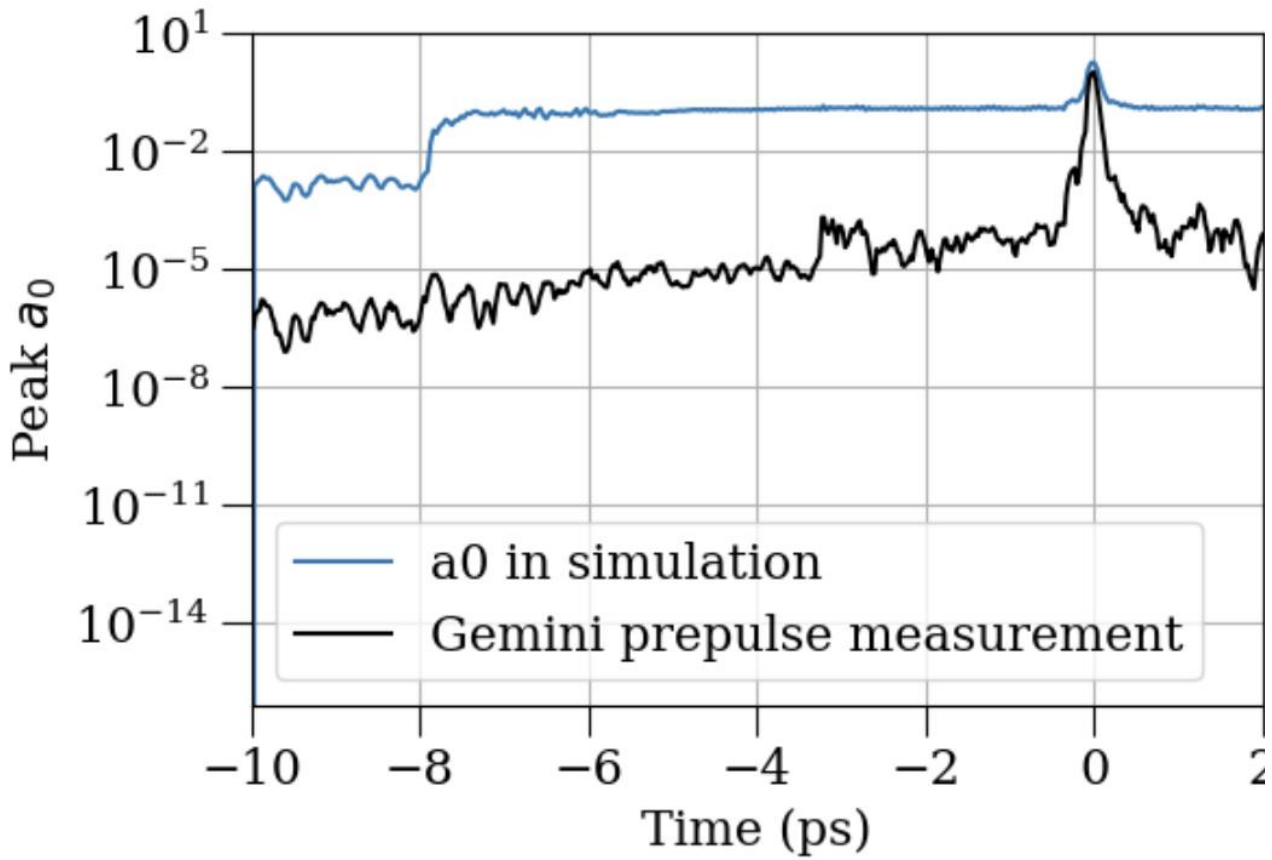
Progress update

Cluster-enhanced injection in LWFA

Prepulse a0 mistake

- Previously used sqrt of the intensity by mistake
- Led to ionization 8ps ahead of main pulse
- Mistake has now been corrected



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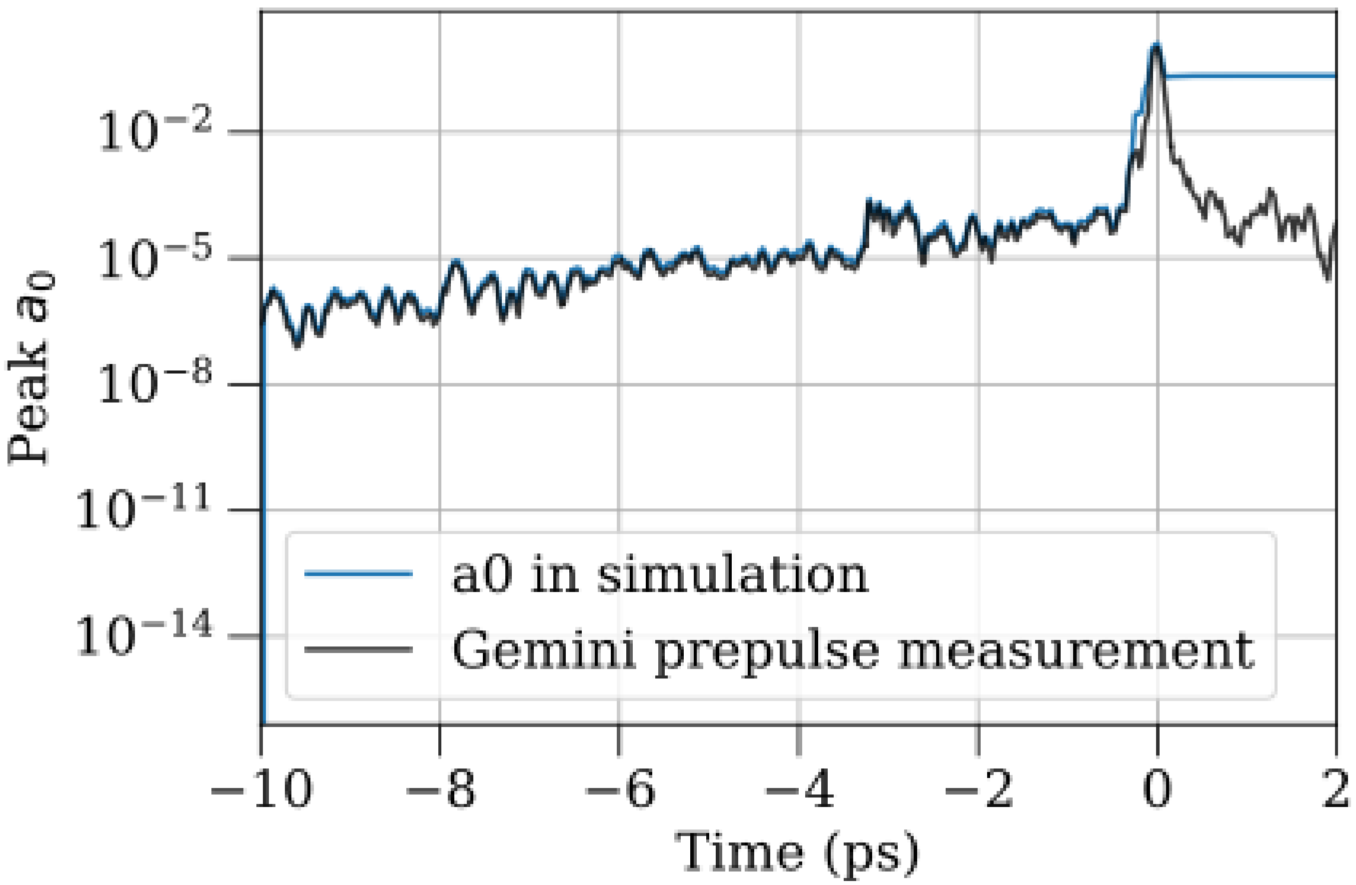
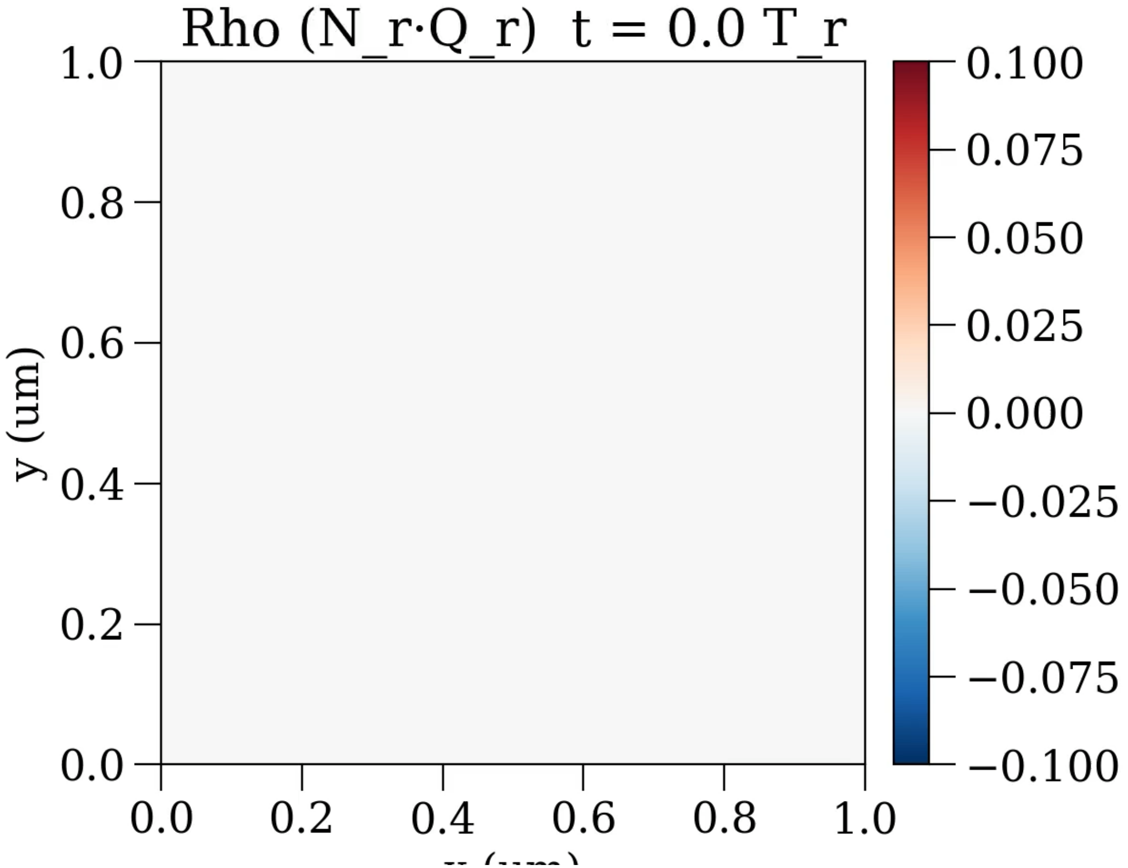
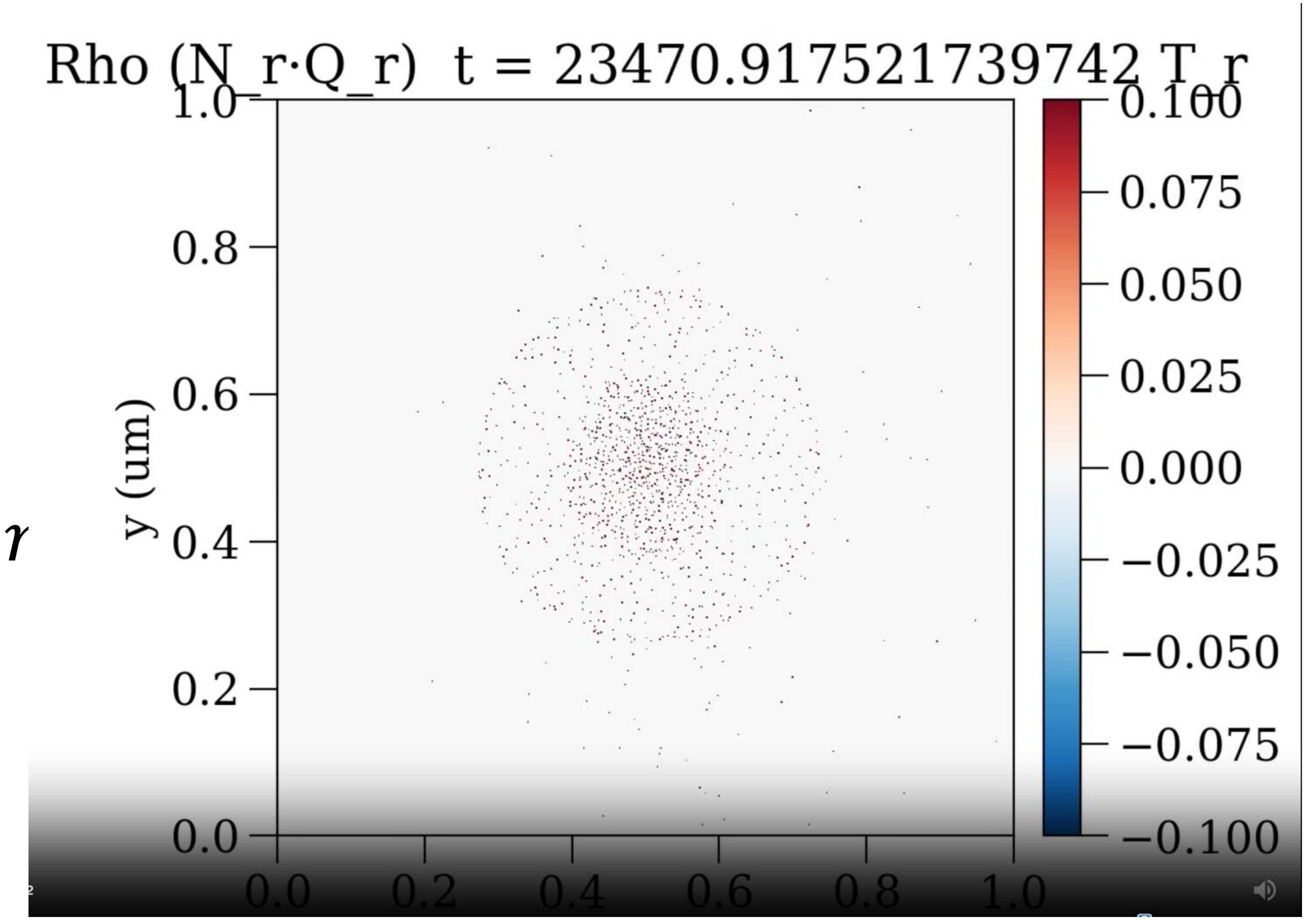


Prepulse expansion of CH4 cluster



Prepulse expansion of CH4 cluster

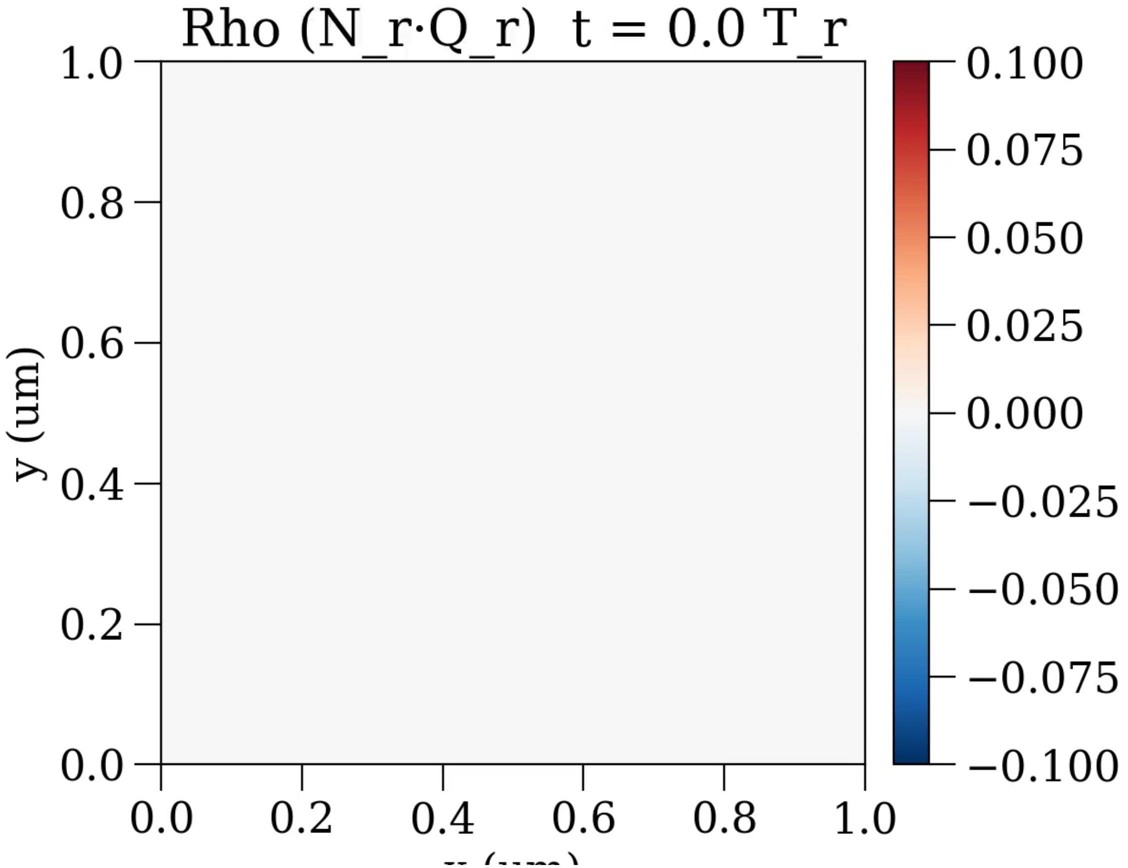
- $\rho = 0.84\, n_c$ when the main pulse arrives
- Ionisation begins to occur 0.25ps ahead of the main pulse
- Cluster parameters: $\rho = 91\, n_c, r$ $10nm$



Liquid CH4 cluster density calculation

- Liquid methane $\rho = 0.42 gcm^{-3}$, molecular mass $M = 16.04\ gmol^{-1}$
- Yields electron number density $n_e = 1.58\ \times\ 10^{23} = 91 n_{crit}$, assuming $n_e = 10 n_{CH4}$
- Cluster radius of $50nm$ in simulations, $V_{cluster} = 5.24\ \times\ 10^{-22}\ m^{-3}$
- This gives $N_{CH4} = n_{CH4} V = 8.3\ \times\ 10^{6}$ molecules
- $N_e = 10\ N_{CH4} = 8.3\ \times\ 10^{7}$ electrons from the fully ionised cluster

Prepulse expansion of CH4 cluster



LWFA baseline with unclustered CH4